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Author(s): Leo A. Goodman

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NOTES ON RESEARCH AND TEACHING



ECOLOGICAL REGRESSIONS AND BEHAVIOR OF INDIVIDUALS*

LEO A. GOODMAN

University of Chicago

In a very interesting paper¹ which was published in 1950, W. S. Robinson pointed out that ecological correlations cannot be used as substitutes for individual correlations. He stated the purpose of his paper will be accomplished "if it prevents the future computation of meaningless correlations and stimulates the study of similar problems with the use of meaningful correlations between the properties of individuals." Since 1950, many papers which base conclusions on ecological correlations have been published. The purpose of this note is to help clarify this problem. We shall emphasize the point that *in general* the study of the regression between ecological variables cannot be used as substitutes for the study of the behavior of individuals. We shall also point out that *in very special circumstances* the study of the regression between ecological variables may be used to make inferences concerning the behavior of individuals.

Why do sociologists study the regression between ecological variables? One answer is that they are interested in the questions which that kind of study can answer. For example, they may be interested in knowing whether or not there is a linear relation between the per cent illiterate and the per cent Negro for the Census Bureau's nine geographic divisions of the United States in 1930. They might want to see whether the geographic divisions which have a high percentage of Negroes also have a high percentage of illiteracy. These kinds of questions are concerned with ecological variables (percentages for geographical districts). The study of the regression between ecological variables (the per cent illiterate and the per cent Negro) is suitable for studying these kinds of questions.

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¹ W. S. Robinson, "Ecological Correlations and Behavior of Individuals," *American Sociological Review*, 15, (June, 1950), pp. 351-357.

Another reason why sociologists study the regression between ecological variables is that they think that such a study may be used to make inferences concerning the behavior of individuals. In one sense these sociologists are incorrect and in another sense they are quite correct. The sociologist *cannot* conclude that the percentage of illiteracy among the Negro population is higher than the percentage of illiteracy among the non-Negro population simply because the correlation between percentage illiteracy and percentage Negro is .946 when the ecological areas are divisions. The percentage of illiteracy among the Negro population might be equal to the percentage of illiteracy among the non-Negro population and we still could obtain an ecological correlation of .946. In fact, the percentage of illiteracy among the Negro population might even be much less than the percentage of illiteracy among the non-Negro population and we still could obtain an ecological correlation of .946. A very high ecological correlation may be obtained when either (a) the individual correlation is close to 0, (b) the individual correlation is close to +1, or (c) the individual correlation is close to -1. Also, a very low ecological correlation may be obtained when either (a), (b), or (c) occur (i.e., when the individual correlation has almost any value at all). Hence, knowledge of the ecological correlations cannot *in general* be used to make inferences concerning the behavior of individuals. Since, however, sociologists continue to use ecological correlations to make such inferences, it is of interest to see in what circumstances these inferences are justified. We shall state the problem somewhat more abstractly than the color-illiteracy study so that the reader will easily see both the generality and the limitations.

Suppose a population can be divided into two mutually exclusive groups, N and W, with respect to one categorization and into two other mutually exclusive groups, I and L, with respect to another categorization. There is an unknown population parameter p , which we shall call the "average probability that a person who is in group N will also be in I." Similarly there is another unknown population parameter r which we call the "average probability that a person who is in group W will also be in I." We shall be in-

interested in ways to estimate these unknown population parameters p and r . If a sample of n people is obtained from N we would expect that np people in this sample would be in I ; i.e., $E\{u\} = np$, where $E\{u\}$ is the expected number of people in I appearing in the sample from N . If a sample of w people is obtained from W , we would expect that wr people in this sample would be in I ; $E\{v\} = wr$, where $E\{v\}$ is the expected number of people in I appearing in the sample from W . If a sample consisting of n people from N and w people from W is obtained, then we would expect that $np + wr$ people in the total sample would be in I ; i.e., $E\{u + v\} = np + wr$, where $E\{u + v\}$ is the expected number of people appearing in I in the total sample. Dividing both sides of this last equation by the total sample size of $n + w$, we find that the expected proportion of people appearing in I in the total sample is $E\{Y\} = Xp + (1 - X)r$, where X is the proportion $n/(n + w)$ of people from N in the total sample. We may rewrite this last equation in the form $E\{Y\} = r + (p - r)X$ or as $E\{Y\} = a + bX$, where $r = a$ and $p - r = b$. Since p and r are unknown population parameters, a and b will also be unknown population parameters. If several samples are obtained which have different values of X (different proportions of people from N in each sample), we might also observe the Y values in each sample (the proportions of people appearing in I in each sample). Plotting the values of Y and X for each sample on a graph gives us the usual scatter diagram (e.g., Figure 1 in Robinson's paper. We then could find the "best fitting" straight line for the scatter diagram by the method of least squares.² In so doing we will also find the least squares estimate for a (the Y intercept) and the least squares estimate for b (the slope). Since $a = r$, we might estimate r by the least squares estimate for a . Also since $a + b = p$, we might estimate p by the sum of the least squares estimates for a and b . These estimates of p and r will be unbiased.

We have seen how to estimate the unknown population parameters p and r using only ecological variables X and Y (the per cent appearing in N in each sample and the per cent in I in each sample). Hence we have used only ecological variables to determine the average probability p that a person who is in group N will also be in I , and the average probability r that a person who is in group W will also be in

I . These average probabilities are relevant to the "individual correlations" discussed by Robinson. Clearly the analysis of fourfold tables in his Table 2 would lead to better estimates of p and r than the estimates we have just obtained by means of a regression analysis of the ecological variables. However, in some cases, where it is too difficult to obtain the data in the cross-classified form of the fourfold tables, the regression method of estimation may be the best approach.

Let us review the assumptions which are necessary in order to carry out the preceding analysis: (a) we wish to estimate two population parameters p and r which do not differ from sample to sample; (b) we have the relation $E\{Y\} = Xp + (1 - X)r$, where X is the proportion of people from N in a sample and $E\{Y\}$ is the expected proportion of people appearing in I in a sample. If (a) and (b) are true, then the standard least squares approach will lead to unbiased estimates of p and r . If, furthermore, we are able to assume that (c) the values of Y are approximately normally distributed with the same variance for each value of X , then all of standard regression methods (test of hypotheses, confidence intervals) may also be applied. The reader is cautioned to examine assumptions (a), (b), and (c) in each particular case before proceeding with a regression analysis of the ecological variables.

We might wish to modify slightly the standard regression methods in order to deal with the special type of problem we have discussed. In our analysis we know that p and r must lie between 0 and 1 since they are "average probabilities." It is possible that both estimates of p and r obtained by the regression approach may not be between 0 and 1. For example, we may find from the ecological data that the regression approach leads to the estimates .2 and -.05 for p and r respectively. If this result was obtained we would be suspicious of our assumptions (a), (b), and (c), although it is possible to obtain such estimates even if these assumptions are satisfied. If these assumptions were satisfied, then we might estimate the value of r as 0 rather than -.05. We would then determine a new least squares estimate for p using the equation $E\{Y\} = Xp$.

In summary, we would like to emphasize that *in general* ecological regression methods cannot be used to make inferences about individual behavior. However, *under very special circumstances* the analysis of the regression between ecological variables may be used to make such inferences.

² W. J. Dixon and F. J. Massey, *Introduction to Statistical Analysis*, New York: 1951, McGraw-Hill Book Company, Inc.